

FIITJEE NSEC MOCK TEST-2

Time Allotted: 2 Hours

CODE:

Maximum Marks: 216

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Write the question paper code (mentioned above) on YOUR OMR Answer Sheet (in the space provided), otherwise your Answer Sheet will NOT be evaluated. Note that the same Question Paper Code appears on each page of the question paper.

Instructions to Candidates:

1. Use of mobile phone, smart watch, and ipad during examination is STRICTLY PROHIBITED.
2. In addition to this question paper, you are given OMR Answer Sheet along with Candidate's copy.
3. On the OMR sheet, make all the entries carefully in the space provided **ONLY** in **BLOCK CAPITALS** as well as by properly darkening the appropriate bubbles.

Incomplete/ incorrect/ carelessly filled information may disqualify your candidature.

4. On the OMR Answer Sheet, use only **BLUE** or **BLACK BALLPOINT PEN** for making entries and filling the bubbles.
5. **Your Enrollment number entered** on the OMR Answer Sheet shall remain your login credentials means login id and password respectively for accessing your performance / result in NSE.
6. Question paper has two parts. In part A1 (Q. No.1 to 48) each question has four alternatives, out of which only one is correct. Choose the correct alternative (s) and fill the appropriate bubble (s), as shown.

Q. No. 22	☐ a	☒	☐ c	☐ d
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In part A2 (Q. No. 49 to 60) each question has four alternatives out of which any number of alternative (s) (1, 2, 3, or 4) may be correct. You have to choose all correct alternative(s) and fill the appropriate bubble(s), as shown

Q. No. 54	☐ a	☒	☐ c	☒
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7. For **Part A1**, each correct answer carries 3 marks whereas 1 mark will be deducted for each wrong answer. In **Part A2**, you get 6 marks if all the correct alternatives are marked. No negative marks in this part.
8. Rough work may be done in the space provided. There are 15 printed pages in this paper
9. Use of **non - programmable scientific** calculator is allowed.
10. No candidate should leave the examination hall before the completion of the examination.
11. After submitting answer paper, take away the question paper & Candidate's copy OMR sheet for your reference.

Please DO NOT make any mark other than filling the appropriate bubbles properly in the space provided on the OMR answer sheet.

OMR answer sheets are evaluated using machine, hence CHANGE OF ENTRY IS NOT ALLOWED. Scratching or overwriting may result in a wrong score.

DO NOT WRITE ON THE BACK SIDE OF THE OMR ANSWER SHEET.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

**INDIAN ASSOCIATION OF PHYSICS TEACHERS
 NATIONAL STANDARD EXAMINATION IN CHEMISTRY
 (NSEC-2023)**

Time: 120 minutes

Max. Marks: 216

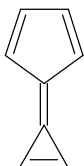
Attempt All Sixty Questions

A – 1

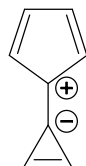
ONLY ONE OUT OF FOUR OPTIONS IS CORRECT BUBBLE THE CORRECTION OPTION

1. Which of the following resonating structure is most unstable?

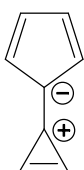
(A)



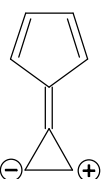
(B)



(C)



(D)



2. An optically pure compound X, gave an $[\alpha]_D^{25} = +20.0^\circ$. A mixture of X and its enantiomer Y, gave an $[\alpha]_D^{25} = +10.0^\circ$. The ratio of X to Y in the mixture is

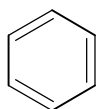
(A) 2 : 1

(B) 1 : 3

(C) 3 : 1

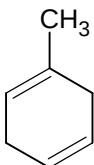
(D) 1 : 2

3. In the following reaction,

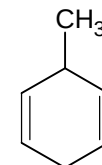


→ Products. One of the final product will be

(A)

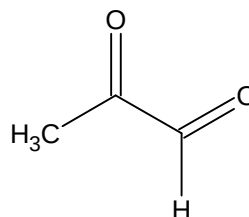


(B)



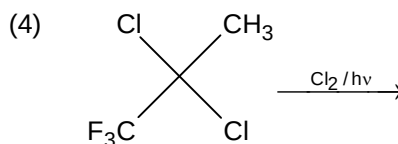
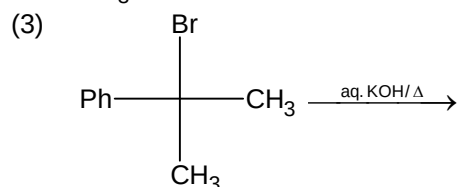
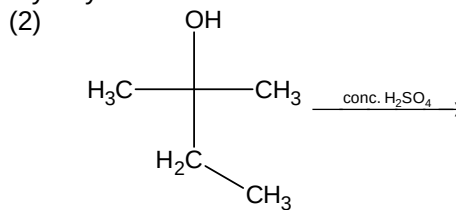
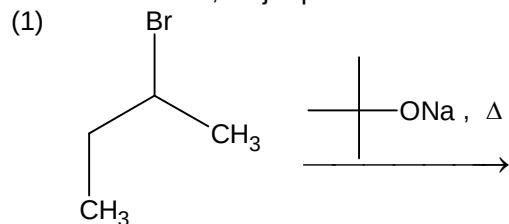
(C) $\text{OHC} - \text{CH}_2 - \text{CHO}$

(D)



Space for rough work

4. In which reaction, major product formation occurs by Saytzeff rule?



Select the correct alternative from the codes given below:

Codes:

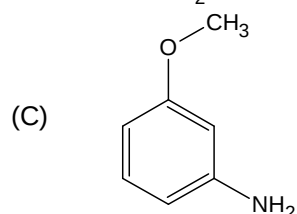
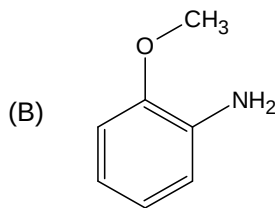
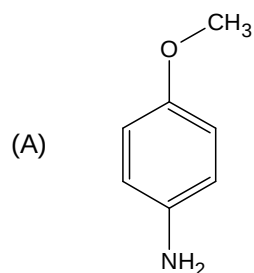
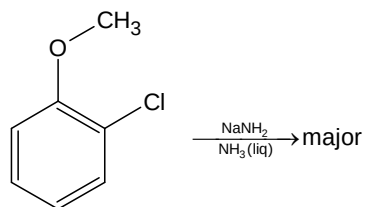
(A) 1, 2 and 3

(B) 2 and 3

(C) 1, 2 and 4

(D) 2, 3 and 4

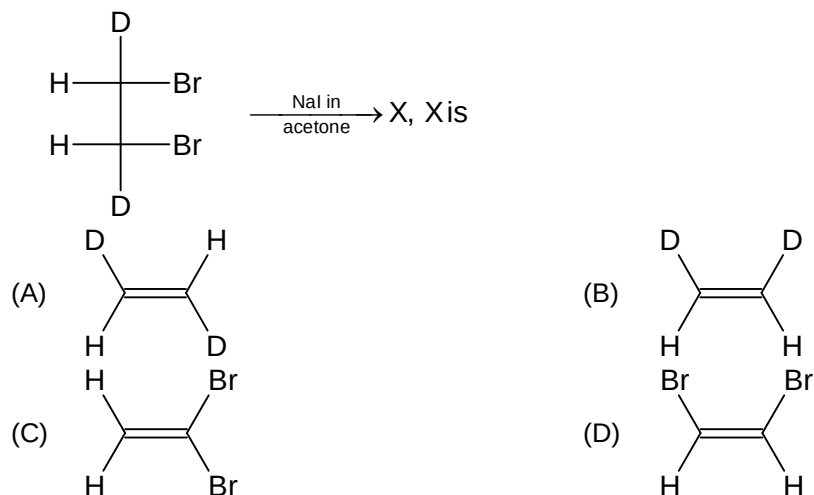
5.



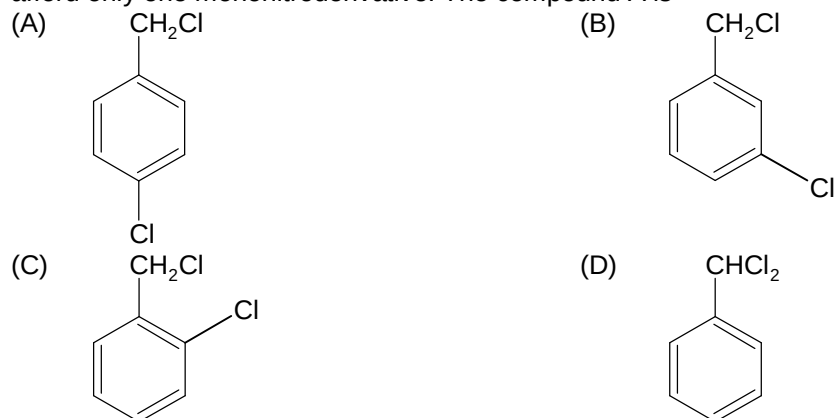
(D) All

Space for rough work

6.

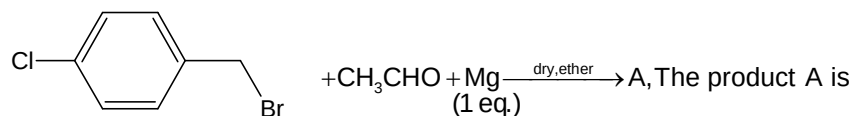


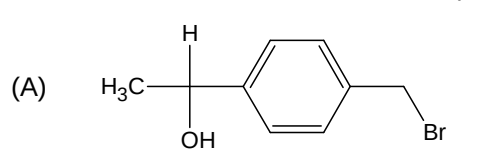
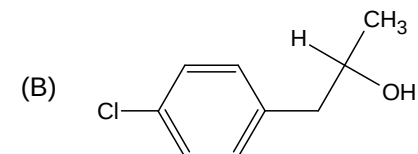
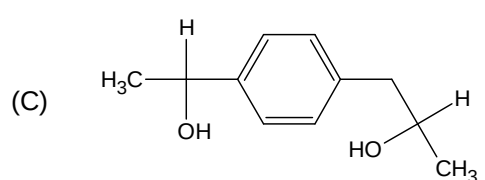
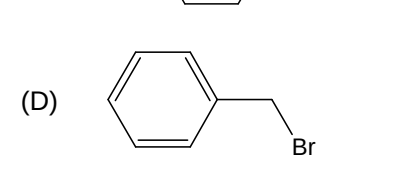
7. An aromatic compound 'A' $\text{C}_7\text{H}_6\text{Cl}_2$ gives white precipitate on boiling with AgNO_3 solution and yields $\text{C}_7\text{H}_7\text{OCl}$ on treatment with NaOH . 'A' on oxidation gives a monochloro benzoic acid which afford only one mononitroderivative. The compound A is

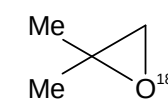


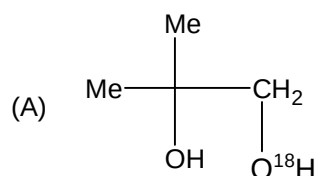
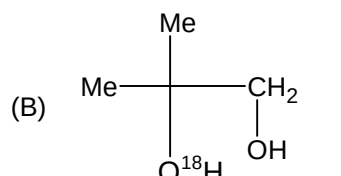
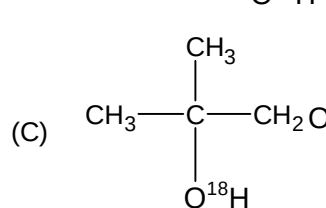
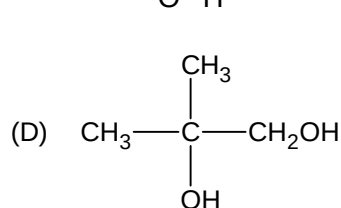
Space for rough work

8. In the reaction



- (A) 
- (B) 
- (C) 
- (D) 

9.  $\xrightarrow[\text{H}_2\text{O}]{\text{H}^+}$ A. A is

- (A) 
- (B) 
- (C) 
- (D) 

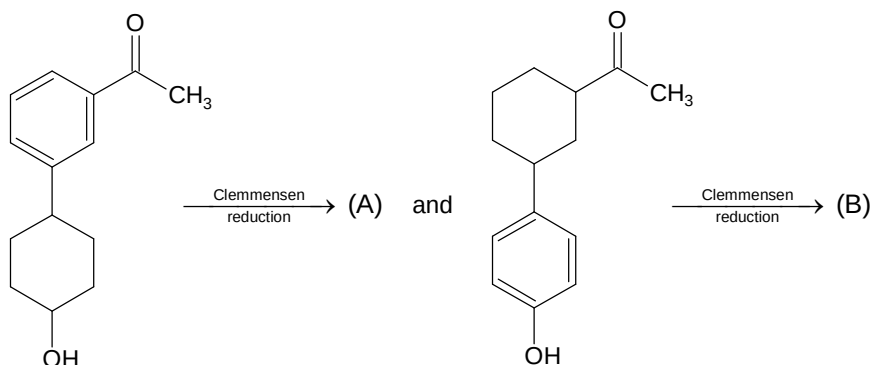
10. The compound exhibits both kinds of stereoisomerism and capable of responding to the iodoform test is

- (A) $\text{Me}-\text{CH}(\text{OH})-\text{CH}=\text{CHMe}$ (B) $\text{MeCH}(\text{OH})-\text{CH}=\text{CHCH}(\text{OH})\text{Me}$
- (C) $\text{MeCH}(\text{OH})-\text{CH}_2-\text{CH}=\text{CH}_2$ (D) None of these

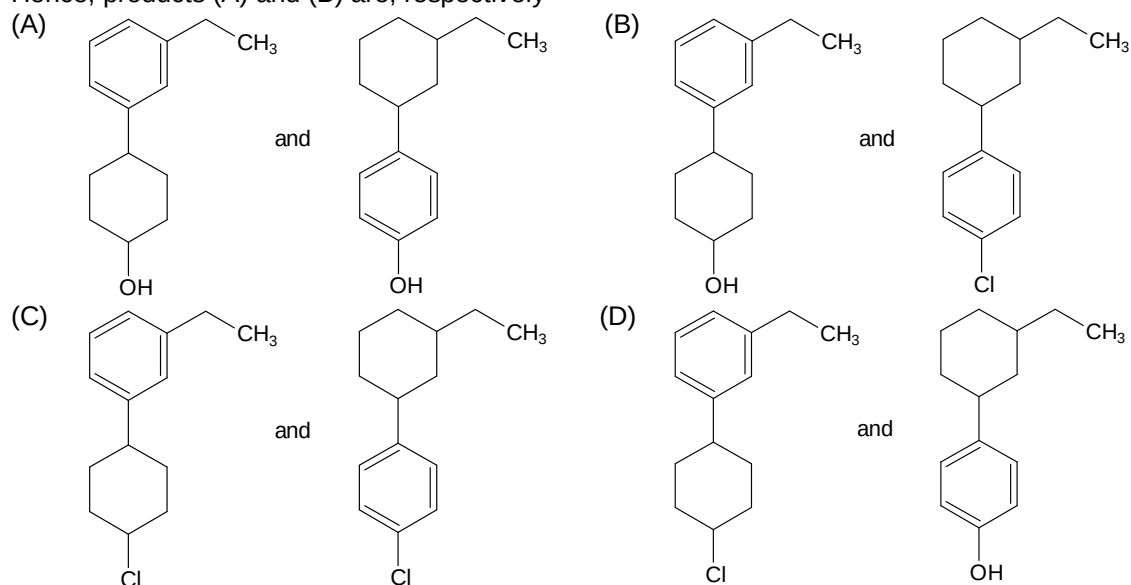
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11. A compound A when reacted with PCl_5 and then with ammonia gave B. B when treated with bromine and caustic potash produced C. C on treatment with NaNO_2 and HCl at 0°C and then boiling produced ortho-cresol. Compound A is
 (A) o-toluic acid (B) o-chlorotoluene
 (C) o-bromotoluene (D) m-toluic acid

12.

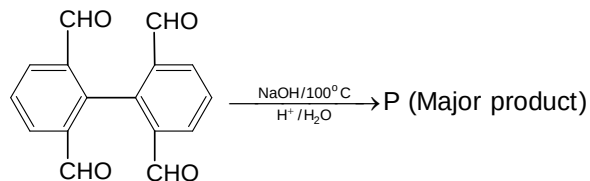


Hence, products (A) and (B) are, respectively

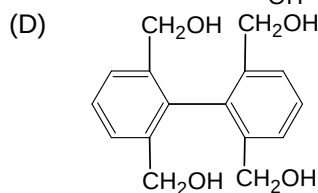
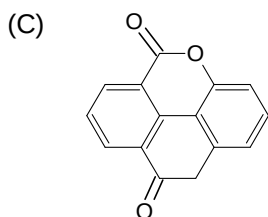
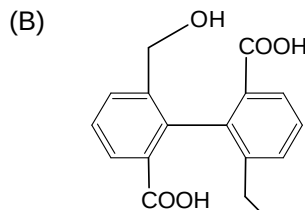
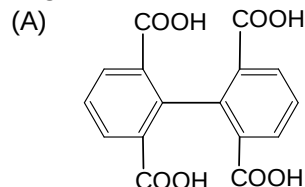


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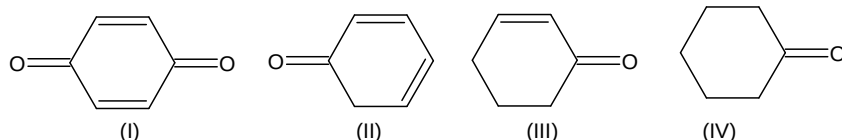
13.



P is



14. Write the following in order of their enol content



(A) $\text{I} < \text{II} < \text{III} < \text{IV}$
 (C) $\text{IV} < \text{I} < \text{III} < \text{II}$

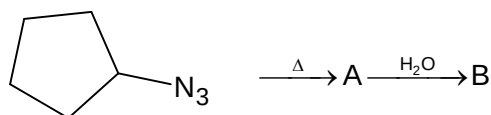
(B) $\text{I} < \text{IV} < \text{III} < \text{II}$
 (D) $\text{II} < \text{III} < \text{IV} < \text{I}$

15. $\text{Br}-\text{CH}_2-\text{CH}_2-\text{Br}$ can be converted into $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{COOH}$ by using

(A) $\text{NaCl}, \text{H}_2\text{O} + \text{H}_2\text{SO}_4$
 (C) $\text{H}_2\text{O} + \text{H}_2\text{SO}_4, \text{KMnO}_4$

(B) $\text{NaCN}, \text{H}_2\text{O} + \text{H}_2\text{SO}_4$
 (D) $\text{KCl}, \text{H}_2\text{O} + \text{H}_2\text{SO}_4$

16.



The product B

(A) Gives positive Tollen's test but does not react with CHCl_3/KOH
 (B) Gives positive isocyanide test but does not perform Tollen's Test
 (C) Does not perform both tests
 (D) Gives positive Tollen's test as well as isocyanide test

Space for rough work

17. Consider the following statements for monosaccharides
(1) They are optically active polyhydroxy carbonyl compound
(2) Fructose is a ketose sugar and hence it does not give red precipitate with Fehling solution
(3) α - D - glucose and β - D - glucose are anomer
Which one of the above statements are corrects
(A) 1 & 2 (B) 1, 2 & 3
(C) 1 and 3 (D) 2 & 3
18. A 0.174 g sample of pyrolusite ore is heated with concentrated HCl when the chlorine evolved is passed through KI solution, it liberates 0.254 g of iodine. The purity of sample is
(A) 40% (B) 50%
(C) 60% (D) 70%
19. 6.90 gm of metal carbonate were dissolved in 60 ml of 2 N HCl. The excess acid was neutralized by 20 ml of 1 N NaOH. What is the equivalent weight of metal?
(A) 40 (B) 20
(C) 19 (D) 39
20. At low pressure van der Waals' equation for 3 moles of a real gas will have its simplified form.
(A) $\frac{PV}{RT + Pb} = 3$ (B) $\frac{PV}{RT - \frac{a}{V}} = 3$
(C) $\frac{PV}{RT + 3Pb} = 1$ (D) $\frac{PV}{RT - \frac{3a}{V}} = 3$
21. A gas is heated in such a way that its pressure and volume both becomes double. Then by decreasing temperature, some of the molecules of air has been taken in container to maintain the doubled volume and pressure. Assuming $\frac{1}{4}$ th of initial number of moles has been taken in for the purpose. By what fraction the temperature must have been raised finally of initial absolute temperature?
(A) 4 times (B) $\frac{16}{5}$ times
(C) $\frac{4}{5}$ times (D) $\frac{1}{5}$ times

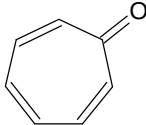
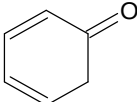
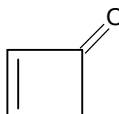
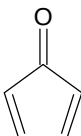
Space for rough work

22. If there were three possible values $\left(-\frac{1}{2}, 0, +\frac{1}{2}\right)$ for the spin quantum number s , the number of elements would be there in the fourth period of periodic table are (assume the other quantum numbers are as usual).
 (A) 18 (B) 27
 (C) 36 (D) 45
23. The frequency of de Broglie wave associated with a microscopic particle of mass 10^{-27} gm is 2.5×10^4 MHz, then the velocity of the particle will be
 (A) 4.08 km/s (B) 5.81 km/s
 (C) 17.32 km/s (D) 1.22 km/s
24. The rate law for the reaction, $2\text{H}_2(\text{g}) + 2\text{NO}(\text{g}) \longrightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$, is experimentally found out to be $\text{rate} = k [\text{H}_2] [\text{NO}]^2$. Which of the following mechanisms can be proposed for the above reaction?
 (I) $2\text{NO} \rightleftharpoons \text{N}_2\text{O}_2$ (fast)
 $\text{N}_2\text{O}_2 + \text{H}_2 \longrightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$ (slow)
 $\text{N}_2\text{O} + \text{H}_2 \longrightarrow \text{N}_2 + \text{H}_2\text{O}$ (fast)
 (II) $\text{H}_2 + \text{NO} \longrightarrow \text{H}_2\text{O} + \frac{1}{2}\text{N}_2$ (slow)
 $\frac{1}{2}\text{N}_2 + \text{NO} \longrightarrow \text{N}_2 + \frac{1}{2}\text{O}_2$ (fast)
 $\frac{1}{2}\text{O}_2 + \text{H}_2 \longrightarrow \text{H}_2\text{O}$ (fast)
 (III) $\text{H}_2 + 2\text{NO} \longrightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$ (slow)
 $\text{N}_2\text{O} + \text{H}_2 \longrightarrow \text{N}_2 + \text{H}_2\text{O}$ (fast)
 (IV) $\text{H}_2 + 2\text{NO} \rightleftharpoons \text{H}_2\text{O} + \text{N}_2\text{O}$ (fast)
 $\text{N}_2\text{O} + \text{H}_2 \longrightarrow \text{N}_2 + \text{H}_2\text{O}$ (slow)
 (A) Only I (B) I and III
 (C) Only III (D) III and IV
25. The rate constant, the activation energy and Arrhenius parameter of a chemical reaction at 25°C are $3 \times 10^{-4} \text{ s}^{-1}$, 104.4 kJ/mol and $6 \times 10^{14} \text{ s}^{-1}$ respectively. The value of the rate constant at $T \rightarrow \infty$ is
 (A) $2 \times 10^{18} \text{ s}^{-1}$ (B) $6 \times 10^{14} \text{ s}^{-1}$
 (C) ∞ (D) $3.6 \times 10^{30} \text{ s}^{-1}$

Space for rough work

26. Solid silver carbonate is decomposed at 120°C according to the given equation,
 $\text{Ag}_2\text{CO}_3(\text{s}) \rightleftharpoons \text{Ag}_2\text{O}(\text{s}) + \text{CO}_2(\text{g})$
If K_p of this equilibrium is 0.0095 bar at 120°C , then the minimum mole % of CO_2 in air which is sufficient to prevent any loss in weight of Ag_2CO_3 , which undergoes drying process at the same temperature is
(A) 0.095% (B) 0.0095%
(C) 0.95% (D) 9.5%
27. Consider the following two gaseous equilibria at a definite temperature.
(i) $\text{AB}(\text{g}) + \frac{1}{2}\text{B}_2(\text{g}) \xrightleftharpoons{K_1} \text{AB}_2(\text{g})$
(ii) $2\text{AB}_2(\text{g}) \xrightleftharpoons{K_2} 2\text{AB}(\text{g}) + \text{B}_2(\text{g})$
If K_1 & K_2 are the equilibrium constants for the reactions (i) & (ii) respectively then the relation between K_1 & K_2 is
(A) $K_1 \cdot K_2 = 1$ (B) $K_1^2 = K_2$
(C) $K_1^2 \cdot K_2 = 1$ (D) $K_1^2 = K_2(0.5)$
28. The conductivity of saturated solution of BaSO_4 is $3.06 \times 10^{-6} \text{ ohm}^{-1}\text{cm}^{-1}$ and its equivalent conductance is $1.53 \text{ ohm}^{-1}\text{cm}^2\text{equiv}^{-1}$. The K_{sp} for BaSO_4 will be
(A) 4×10^{-12} (B) 2.5×10^{-9}
(C) 2.5×10^{-13} (D) 4×10^{-6}
29. The basic ionisation constant for hydrazine, N_2H_4 is 9.6×10^{-7} . What would be the percent hydrolysis of 0.1 $\text{N}_2\text{H}_5\text{Cl}$?
(A) 0.016% (B) 3.2%
(C) 1.6% (D) 0.032%
30. A cell was prepared by using the electrodes of same electrolyte having same conc. 0.1 M. At what temperature one of these electrodes should be placed so as to get an electrochemical cell of potential difference 0.14, if the other one is present at 27°C . (Take $n = 2$)
(A) These electrodes will not form a cell (B) 719°C
(C) 1438°C (D) -27°C
31. A cell made up of two hydrogen electrodes. The positive electrode is in contact with a solution of 10^{-4} M hydrogen ions. The emf of the cell is 0.118 V at 25°C . The concentration of hydrogen ions at negative electrode is
(A) 10^{-6} M (B) 10^{-4} M
(C) 10^{-2} M (D) None of the above

Space for rough work

32. Find ΔH at 358 K for the reaction,
 $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{Fe}(\text{s}) + 3\text{H}_2\text{O}(\ell)$
 Given that, $\Delta H_{298} = -33.29 \text{ kJ/mole}$ and C_p for $\text{Fe}_2\text{O}_3(\text{s})$, $\text{Fe}(\text{s})$, $\text{H}_2\text{O}(\ell)$ and $\text{H}_2(\text{g})$ are 103.8, 25.1, 75.3 and 28.8 J/K mole.
 (A) 85.9 kJ/mole (B) -28.136 kJ/mole
 (C) +28.136 kJ/mole (D) 85.9 J/mole
33. Which of the following equation is not correct?
 (A) $\Delta G = -T\Delta S_{(\text{Total})}$ (B) $\Delta S = - \left(\frac{d(\Delta G)}{dT} \right)_P$
 (C) $\frac{d \ln K_c}{dT} = \frac{\Delta U}{RT^2}$ (D) All equation are correct
34. Which of the following statements is correct for CsBr_3 ?
 (A) It is a covalent compound
 (B) It contains Cs^{3+} and Br^-
 (C) It contains Cs and Br_3^-
 (D) It contains Cs^+ , Br^- and lattice Br_2 molecule
35. The compound, which has highest dipole moment?
 (A)  (B) 
 (C)  (D) 
36. Among KO_2 , AlO_2^- , BaO_2 and NO_2^+ , unpaired electrons are present in
 (A) NO_2^+ and BaO_2 (B) KO_2 and AlO_2^-
 (C) KO_2 (D) BaO_2

Space for rough work

37. The reason for the difference in the sequence of energy levels of the molecular orbitals for nitrogen and oxygen is that
(A) nitrogen is polar and oxygen is non-polar
(B) in nitrogen, the energy difference between $\sigma 2p_z$ and $\sigma 2s$ is larger than in oxygen
(C) in nitrogen, $\sigma 2s$ and $\sigma 2p_z$ orbitals interact while in oxygen they do not interact
(D) in oxygen, $\sigma 2s$ and $\sigma 2p_z$ orbitals interact
38. The ratio of van't Hoff factor, if PBr_5 and PCl_5 are assumed to be soluble in water and remains same as in solid state, is
(A) 2 (B) 1
(C) 1.5 (D) 2.5
39. Which one of the following statements about the reaction below is false?
 $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + \text{EDTA}^{4-} \rightleftharpoons [\text{Cu}(\text{EDTA})]^{2-} + 6\text{H}_2\text{O}$
(A) $[\text{Cu}(\text{EDTA})]^{2-}$ is a more stable complex than $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$.
(B) Both $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cu}(\text{EDTA})]^{2-}$ are octahedral complexes.
(C) There is an increase in entropy when the reaction occurs.
(D) This is a redox reaction.
40. Among TiF_6^{2-} , CoF_6^{3-} , Cu_2Cl_2 and NiCl_4^{2-} (atomic numbers of Ti = 22, Co = 27, Cu = 29, Ni = 28), the colourless species are
(A) CoF_6^{3-} and NiCl_4^{2-} (B) TiF_6^{2-} and CoF_6^{3-}
(C) Cu_2Cl_2 and NiCl_4^{2-} (D) TiF_6^{3-} and Cu_2Cl_2
41. The correct structure of the $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{4-}$ & $[\text{Ni}(\text{Cl})_4]^{2-}$ are
(A) Tetrahedral, tetrahedral & tetrahedral respectively.
(B) Tetrahedral, square planar & square planar respectively.
(C) Square planar, square planar & square planar respectively.
(D) Tetrahedral, square planar & tetrahedral respectively.
42. The slag obtained during the extraction of copper pyrites is composed mainly of
(A) Cu_2S (B) FeSiO_3
(C) CuSiO_3 (D) SiO_2
43. An inorganic Lewis acid (X) fumes in moist air and intensity of fumes increases when a rod dipped in NH_4OH is brought near to it. An acidic solution of (X) on addition of NH_4Cl and NH_4OH gives a precipitate which dissolves in NaOH solution. An acidic solution of (X) does not give precipitate with H_2S . Hence, the compound (X) is
(A) FeCl_3 (B) AlCl_3
(C) SnCl_2 (D) ZnCl_2

Space for rough work

44. The hydroxide which has lowest value of K_{sp} at 25°C is
 (A) $\text{Be}(\text{OH})_2$ (B) $\text{Ba}(\text{OH})_2$
 (C) $\text{Ca}(\text{OH})_2$ (D) $\text{Mg}(\text{OH})_2$
45.
$$\begin{array}{ccc} \text{H}_3\text{PO}_4 & \xrightarrow{250^\circ\text{C}} & (\text{X}) \\ & \searrow 600^\circ\text{C} & \downarrow 600^\circ\text{C} \\ & & (\text{Y}) \end{array}$$

 (A) $\text{X} - \text{H}_4\text{P}_2\text{O}_7(\ell)$
 $\text{Y} - \text{HPO}_3(\text{s})$
 (B) $\text{X} - \text{H}_4\text{P}_2\text{O}_7(\text{s})$
 $\text{Y} - \text{PH}_3(\text{g})$
 (C) $\text{X} - \text{H}_4\text{P}_2\text{O}_7(\text{s})$
 $\text{Y} - \text{P}_2\text{O}_5(\text{g})$
 (D) $\text{X} - \text{H}_4\text{P}_2\text{O}_7(\ell)$
 $\text{Y} - \text{H}_3\text{PO}_3(\ell)$
46. A compound (A) forms an unstable pale blue colour solution in water. The solution decolourizes Br_2 water and an acidified solution of KMnO_4 . The compound (A) is
 (A) HNO_2 (B) HNO_3
 (C) N_2O_5 (D) None of the above
47. Which pair is amphoteric?
 (i) $\text{Zn}(\text{NH}_2)_2$ (ii) $\text{Al}(\text{OH})_3$
 (iii) $\text{Zn}(\text{OH})_2$ (iv) $\text{Be}(\text{OH})_2$
 (A) (ii), (iii), (iv) (B) (i), (ii), (iii)
 (C) (i), (ii), (iii), (iv) (D) (ii), (iv)
48. Anhydrous borax ($\text{Na}_2\text{B}_4\text{O}_7$) is converted in to B_2O_3 by following steps

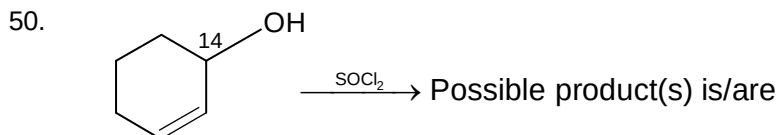
$$\text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\text{acid}} \text{H}_3\text{BO}_3 \xrightarrow{\Delta} \text{B}_2\text{O}_3$$

 If 10.1 gm of impure anhydrous borax sample is being treated with acid followed by heating, the amount of B_2O_3 produced 3.5 gm. Find the percentage purity of borax sample
 (A) 100 % pure (B) 50 % pure
 (C) 25 % pure (D) 12.5 % pure

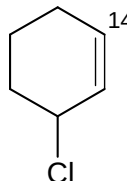
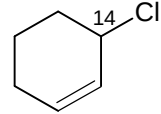
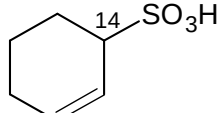
Space for rough work

ANY NUMBER OF OPTIONS 4, 3, 2 or 1 MAY BE CORRECT
 MARKS WILL BE AWARDED ONLY IF ALL THE CORRECT OPTIONS ARE BUBBLED

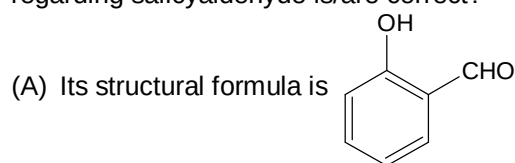
49. Which of the following statements is/are correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4 ?
 (A) A deep red vapour is evolved.
 (B) The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4 .
 (C) Chlorine gas is evolved.
 (D) Chloromyl chloride is formed.



Which of the following options is major product?

- (A)  (B) 
 (C)  (D) None of the above

51. Phenol gives salicylaldehyde on heating with $CHCl_3$ and NaOH. Which of the following statement(s) regarding salicylaldehyde is/are correct?



- (B) It is steam volatile.
 (C) Phenolic group is not so reactive due to intramolecular hydrogen bonding.
 (D) An aqueous solution of salicylaldehyde gives a violet colouration with $FeCl_3$.

Space for rough work

52. $C_2H_4Cl_2 \xrightarrow{KCN} (B) \xrightarrow{H_3O^+} (C) \xrightarrow{\Delta} (D)$
 (A) The compound D has the characteristic(s)
 (A) It is succinic anhydride if (A) is vic dihalide (B) It is propionic acid if (A) is gem dihalide
 (C) It is succinic anhydride if (A) is gem dihalide (D) It is propionic acid if (A) is vic dihalide
53. Which compounds do not give Lassaigne test for nitrogen?
 (A) $C_6H_5N_2Cl$ (B) NH_2-CH_2-COOH
 (C) NH_2OH (D) CH_3CONH_2
54. The dissociation of phosgene, which occurs according to the reaction $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$ is an endothermic process. Which of the following factors will increase the degree of dissociation of $COCl_2$?
 (A) Adding Cl_2 to the system
 (B) Adding helium to the system at constant pressure
 (C) Decreasing the temperature of the system
 (D) Reducing the total pressure
55. Which of the following statements are correct about half-life period?
 (A) It is proportional to the initial concentration for zero order.
 (B) Mean life is 1.44 times the half-life for the first order reaction.
 (C) 99.9% reaction takes place in 10 minutes only when the rate constant is 0.693 min^{-1} for first order reaction.
 (D) Half-life depends on initial concentration for first order reaction.
56. Which of the following statement(s) is/are correct?
 (A) Oxidation number of Fe in $Na_2[Fe(CN)_5(NO)]$ is +2.
 (B) $[Ag(NH_3)_2]^+$ is linear in shape.
 (C) In $[Fe(H_2O)_6]^{3+}$, Fe is d^2sp^3 hybridized.
 (D) In $[Ni(CO)_4]$, oxidation number of Ni is zero.
57. At a very low pressure and critical temperature, the molar volume of N_2 gas is 0.208 L. The compression-factor of N_2 gas at the above condition will not be (assume N_2 to be van der Waals' gas, $b = 0.037 \text{ L mol}^{-1}$ = van der Waals' constant)
 (A) 0.4 (B) 4.8
 (C) 0.48 (D) 1.125

Space for rough work

58. Which of the following is/are not correct for benzoic acid?
(A) It is more acidic than p-toluic acid.
(B) It is less acidic than o-toluic acid.
(C) It is more acidic than m-methoxy benzoic acid.
(D) It is less acidic than o-methoxy benzoic acid.
59. Which of the following statement(s) is/are true?
(A) A weak base, B, has $K_b = 2.0 \times 10^{-5}$. The pH of any solution in which $[B^-] = [BH]$ is 4.7.
(B) A buffer of acetic acid and potassium acetate in which $[CH_3COOH] = [CH_3COO^-]$ will have $pH = pK_a$.
(C) A mixture of CH_3COOH and NH_4OH acts as a buffer.
(D) The pH of a solution made by mixing 1 M HCl and 1 M NaOH solution is 7.
60. Select incorrect statement(s) about the following:
- $$\begin{array}{c} H & & H \\ & \backslash & / \\ & C = C = C \\ & / & \backslash \\ Cl & & Cl \end{array}$$

(I)

$$\begin{array}{c} H & & CH_3 \\ & \backslash & / \\ & C = C = C \\ & / & \backslash \\ Cl & & CH_3 \end{array}$$

(II)
- (A) Both lack chiral carbons but I is chiral molecule and II is achiral molecule.
(B) Both lack chiral carbons and both are chiral molecules.
(C) Both lack chiral carbons and both are achiral molecules.
(D) Both have chiral carbons and both are chiral molecules.

Space for rough work

ANSWERS

- | | | | |
|-------------|----------|----------------|-------------|
| 1. B | 2. C | 3. C | 4. B |
| 5. C | 6. A | 7. A | 8. B |
| 9. A | 10. A | 11. A | 12. D |
| 13. B | 14. B | 15. B | 16. D |
| 17. C | 18. B | 19. D | 20. D |
| 21. B | 22. B | 23. A | 24. B |
| 25. B | 26. C | 27. C | 28. D |
| 29. D | 30. B | 31. A | 32. B |
| 33. D | 34. C | 35. A | 36. C |
| 37. C | 38. A | 39. D | 40. D |
| 41. A | 42. B | 43. B | 44. A |
| 45. A | 46. A | 47. C | 48. B |
| 49. A, B, D | 50. A, B | 51. A, B, C, D | 52. A, B |
| 53. A, C | 54. B, D | 55. A, B, C | 56. A, B, D |
| 57. B, C, D | 58. C | 59. B, C, D | 60. B, C, D |

HINTS AND SOLUTIONS

1. **B**

Structure (B) is anti-aromatic compound.

2. **C**

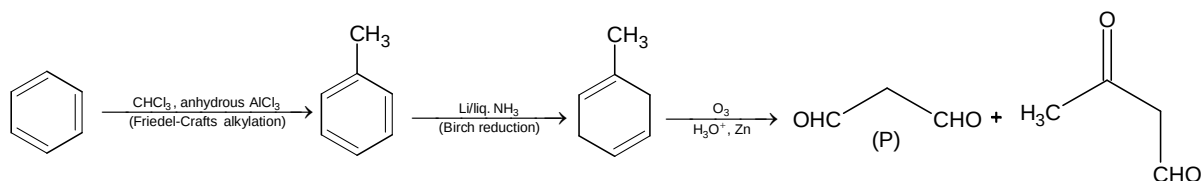
$$\text{Amount of } X = \frac{x}{x+y}, \text{ amount of } Y = \frac{y}{x+y}$$

$$\therefore \text{Specific rotation of } X = 20^\circ \Rightarrow \frac{x}{x+y} \times 20 - \frac{y}{x+y} \times 20 = 10$$

$$\therefore x = 3y$$

$$\therefore x : y = 3 : 1$$

3. **C**

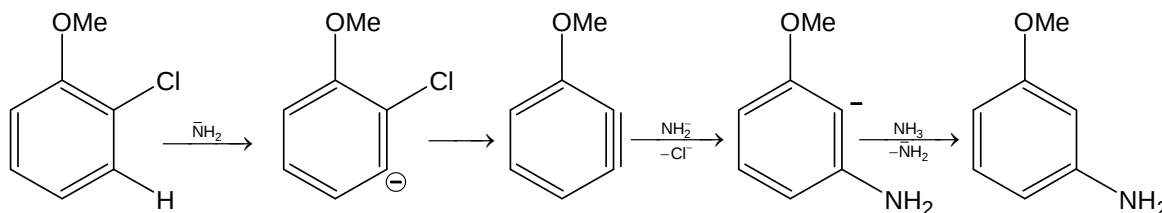


4. **B**

(1) proceeds through E2 path while (2) and (3) proceeds through E1 path and (4) proceeds through E1CB path.

Hence, only 2 and 3 takes place through Saytzeff rule.

5. **C**

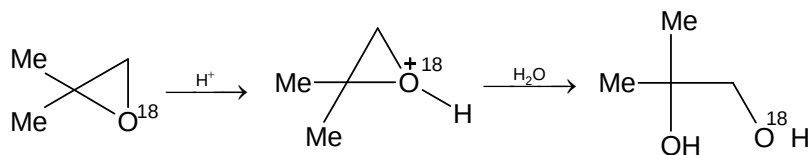


6. **A**

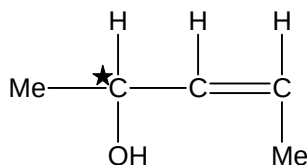
7. **A**

8. **B**

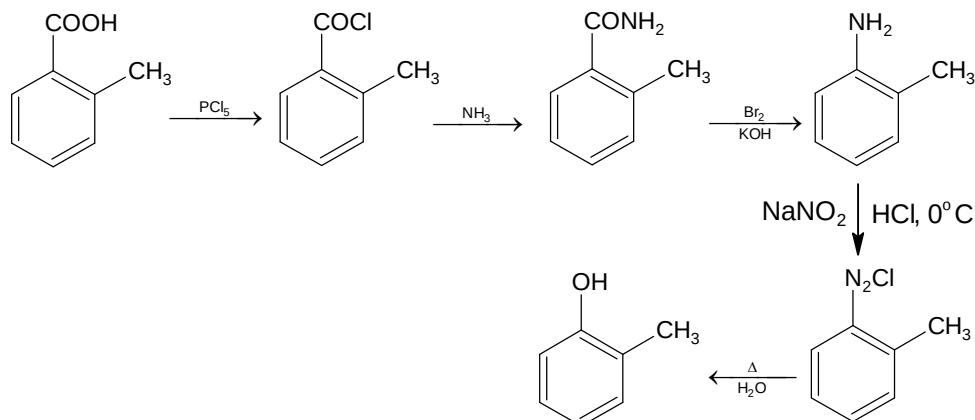
9. **A**



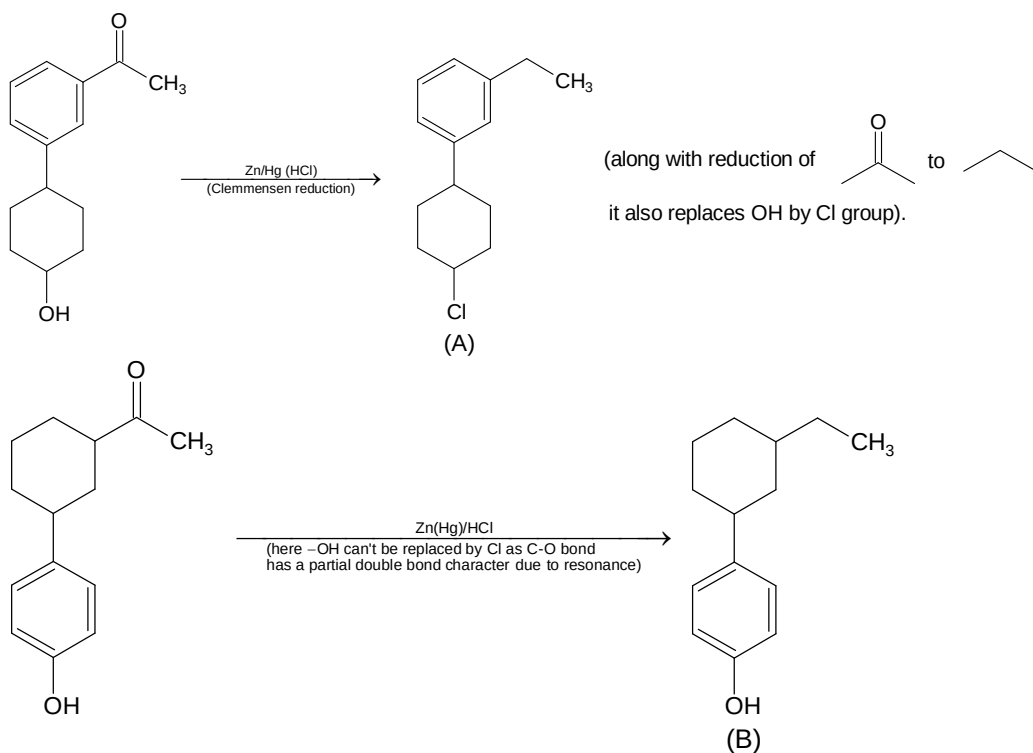
10. **A**



11. **A**



12. **D**

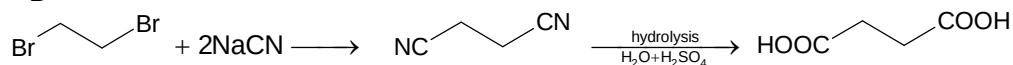


Hence, (D).

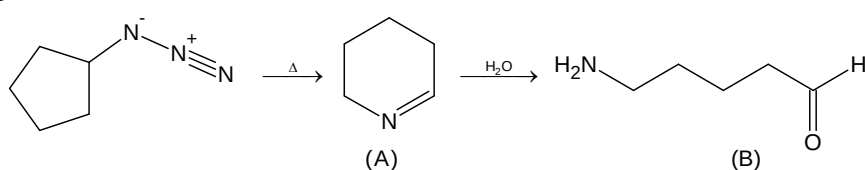
13. **B**

14. **B**

15. **B**



16. **D**



As B compound contains both an aldehyde group as well as P amine have it give both tests.

17. **C**

18. **B**

19.

D

Equivalent of HCl taken = $60 \times 2 \times 10^{-3}$

Equivalent of HCl present after the reaction = $20 \times 1 \times 10^{-3}$

$\therefore$ Equivalent of HCl utilized = 100×10^{-3}

$\therefore 100 \times 10^{-3}$ equivalent of metal carbonate = 6.90 gm

$\therefore 1$ equivalent of metal carbonate = $\frac{6.90}{10^{-1}} = 69$ gm

Equivalent weight of metal = $69 - 30 = 39$

20.

D

21.

B

For first step:

$$\frac{P_1 V_1}{T_1} = \frac{2P_1 \times 2V_1}{T_2} \Rightarrow T_2 = 4T_1$$

By doing second step (an open system), i.e. P and V will remain constant.

$$\text{or, } 4T_1 \times n_1 = T'_2 \times n'_2$$

$$\text{or, } 4T_1 n_1 = T'_2 \times \frac{5}{4} n_1 \left(\text{as } n'_2 = n_1 + \frac{n_1}{4} \right)$$

$$\therefore T'_2 = \frac{16}{5} T_1$$

22.

B

23.

A

24.

B

25.

B

26.

C

$$K_p = P_{\text{CO}_2} = 0.0095 \text{ bar} = 9.5 \times 10^{-3} \text{ bar}$$

If partial pressure of CO_2 in air becomes equal to 9.5×10^{-3} bar or greater than 9.5×10^{-3} bar, then decomposition of Ag_2CO_3 in drying process is restricted, i.e. no weight loss in Ag_2CO_3 during heating in this condition.

$$\begin{aligned} \text{Hence, min. \% of CO}_2 \text{ in air} &= \frac{9.5 \times 10^{-3} \text{ bar}}{1 \text{ bar}} \times 100 \\ &= 0.95\% \end{aligned}$$

27.

C

$$K_1 = \frac{[\text{AB}_2]}{[\text{AB}][\text{B}_2]^{1/2}}, K_2 = \frac{[\text{AB}_2][\text{B}_2]}{[\text{AB}_2]^2}$$

$$K_1^2 \cdot K_2 = \frac{[\text{AB}_2]^2}{[\text{AB}]^2 [\text{B}_2]} \cdot \frac{[\text{AB}]^2 [\text{B}_2]}{[\text{AB}_2]^2}$$

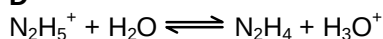
$$K_1^2 \cdot K_2 = 1$$

28.

D

29.

D



$$K_h = \frac{[\text{N}_2\text{H}_4][\text{H}_3\text{O}^+]}{[\text{N}_2\text{H}_5]} = \frac{k_w}{k_b} = \frac{10^{-14}}{9.6 \times 10^{-7}} = \frac{x^2}{0.1} = 1.04 \times 10^{-8}$$

$$\therefore x = 3.2 \times 10^{-5}$$

$$\% \text{ hydrolysis} = \frac{3.2 \times 10^{-5}}{0.1} \times 100 = 0.032\%$$

30.

B

$$E_c = E_c^\circ + \frac{2.303RT_1}{nF} \log[C]$$

$$E_a = E_c^\circ + \frac{2.303RT_2}{nF} \log[C]$$

$$E_c - E_a = + \frac{2.303R(T_1 - T_2)}{nF} \log[C]$$

$$\frac{0.14 \times 2 \times 96500}{2.303R \log C} = T_1 - T_2$$

$$-1411.1 = T_1 - T_2$$

$$T_2 = T_1 + 1411.1$$

31.

A



$$1 \text{ atm} \times 10^{-4} \text{ 1 atm}$$

$$E_{\text{cell}} = \frac{0.0591}{2} \log \frac{(10^{-4})^2}{x^2}$$

$$x = 10^{-6} \text{ M}$$

32.

B

$$\Delta C_p = 2 \times 25.1 + 3 \times 75.3 - (103.8 + 3 \times 28.8) = 85.9 \text{ J/mole}$$

$$\frac{\Delta H_{358} - \Delta H_{298}}{T_2 - T_1} = \Delta C_p$$

$$\Delta H_{358} = -28.136 \text{ kJ/mole}$$

33.

D

34.

C

35.

A

36.

C

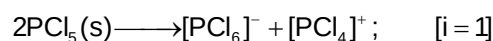
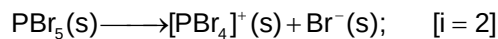
37.

C

Due to interaction, the energy of $\sigma 2p_z$ is higher than $\pi 2p_x$ and $\pi 2p_y$ in nitrogen. On the other hand in oxygen, the interaction is absent, so $\sigma 2p_z$ is lower in energy than $\pi 2p_x$ and $\pi 2p_y$.

38.

A



$$\text{ratio} = \frac{2}{1} = 2$$

39.

D

40.

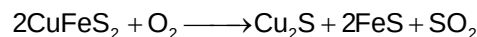
D

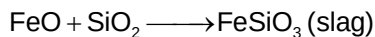
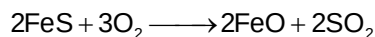
41.

A

42.

B

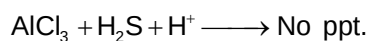
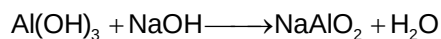
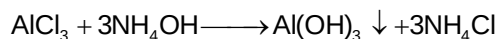
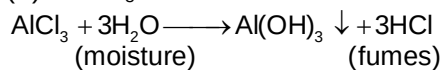




43.

B

(X) is AlCl_3 .



Hence, (B).

44.

A

Solubility of hydroxides increases down the group, hence Be(OH)_2 is least soluble. Hence, least K_{sp} .

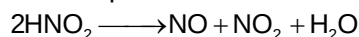
45.

A

46.

A

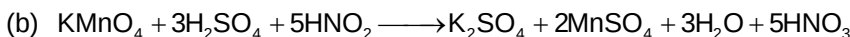
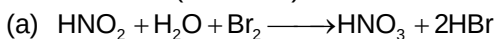
HNO_2 is pale blue colour due to dissociation of N_2O_5 in concentrated solution. It decomposes as



and in dilute solution it decomposes as



Reaction with (Br_2 water)



47.

C

$\text{Zn(NH}_2)_2$ is amphoteric in NH_3 .

Rest are amphoteric in H_2O .

48.

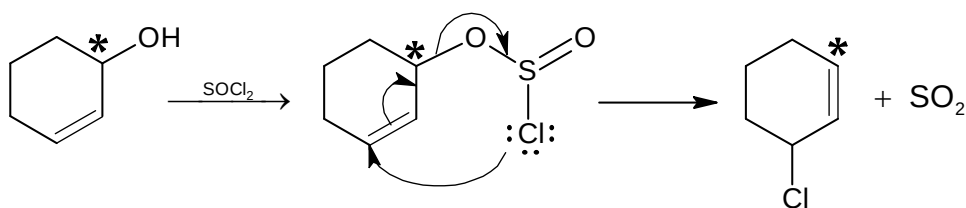
B

49.

A, B, D

50.

A, B

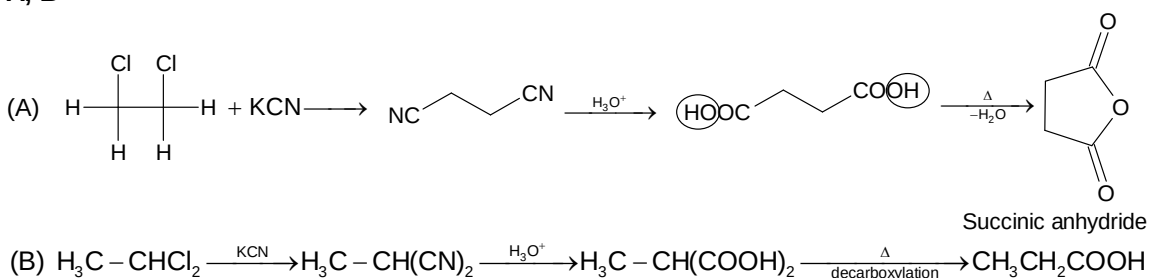


51.

A, B, C, D

52.

A, B



53. **A, C**
(A) Because at the required temperature of Lassaigne test, N_2 expels out in the form of a gas in $C_6H_5N_2Cl$.
(C) NH_2OH does not contain carbon.
54. **B, D**
55. **A, B, C**
56. **A, B, D**
57. **B, C, D**
van der Waals' equation is
$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

where, V_m = molar volume
At low pressure, $V_m \gg b$, then $\left(P + \frac{a}{V_m^2}\right)V_m = RT$
$$\therefore PV_m + \frac{a}{V_m} = RT \Rightarrow PV_m = RT - \frac{a}{V_m}$$

$$\therefore Z = \frac{PV_m}{RT} = 1 - \frac{a}{RTV_m}$$

Now critical temperature, $T_c = \frac{8a}{27Rb}$
$$\therefore \frac{a}{T_c} = \frac{27Rb}{8} = 3.375 Rb$$

 $\therefore$ At low pressure and critical temperature, compressibility factor (Z) is
$$Z = 1 - \frac{a}{T_c R V_m} = 1 - 3.375 Rb \times \frac{1}{R V_m} = 1 - \frac{3.375b}{V_m}$$

$$= 1 - \frac{3.375 \times 0.037}{0.208} = 1 - 0.6 = 0.4$$
58. **C**
59. **B, C, D**
60. **B, C, D**